

8 Python Solver Validation

To validate the python solver, a model of a large plate with an unloaded hole is used. This plate has tensile loading applied at the right edge. The plate is constrained such that all nodes except the center line node along the left edge are free to displace in the y direction. Thus, the structure will not resist the poisons effect due to the applied tension load. The center line node along the left edge is fully fixed to eliminate rigid body motion of the full model. This model can be see in Figure 14.

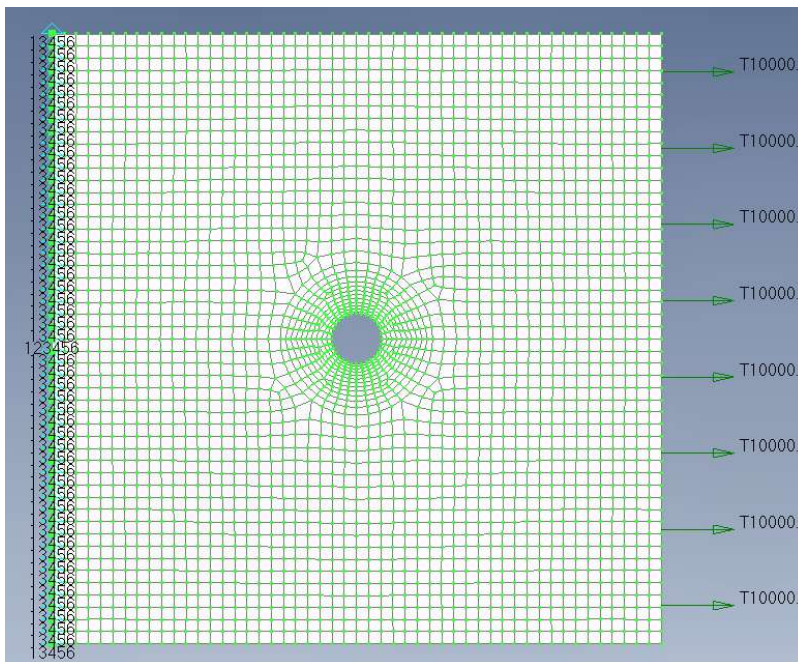


Figure 14: Tension Plate

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8.1 Elasticity Solution

The above model is large enough to assume the same behavior as the classical elasticity solution for the infinite plate with an unloaded hole as defined in the following equations

$$\sigma_{rr} = \frac{\sigma}{2} \left(1 - \frac{a^2}{r^2} \right) + \frac{\sigma}{2} \left(1 + 3 \frac{a^4}{r^4} - 4 \frac{a^2}{r^2} \right) \cos 2\theta$$

$$\sigma_{\theta\theta} = \frac{\sigma}{2} \left(1 + \frac{a^2}{r^2} \right) - \frac{\sigma}{2} \left(1 + 3 \frac{a^4}{r^4} \right) \cos 2\theta$$

$$\sigma_{r\theta} = -\frac{\sigma}{2} \left(1 - 3 \frac{a^4}{r^4} + 2 \frac{a^2}{r^2} \right) \sin 2\theta$$

The critical stress occurs when $r = a$ and $\theta = \frac{\pi}{2}$

$$\sigma_{rr} = \frac{\sigma}{2} \left(1 - \frac{1}{1} \right) + \frac{\sigma}{2} (1 + 3 - 4) \cos \pi = 0$$

$$\sigma_{\theta\theta} = \frac{\sigma}{2} (1 + 1) - \frac{\sigma}{2} (1 + 3) \cos \pi = \sigma - (-2\sigma) = 3\sigma$$

$$\sigma_{r\theta} = -\frac{\sigma}{2} (1 - 3 + 2) \sin \pi = 0$$

This closed form solution provides the exact solution to this problem and shows the familiar Kt of 3 which occurs at $\theta \pm \pi/2$. This an exact solution to which FEM solvers can be compared against to determine validity.

8.2 Femap Solution

Creating this model within FEMAP with a highly refined CQ4 mesh produces the expected max principal stress distribution as shown in Figure 15. This model has 10000 lbf load applied and an area of 1.5in². Calculating the stresses below shows that FEMAP is matching the exact elasticity solution as is expected of a commercially available FEM suite.

$$\sigma_{gross} = \frac{1000 \text{ lbf}}{6in * 0.25in} = 6.67 \text{ ksi}$$

$$\sigma_{ma} = 20.086 \text{ ksi}$$

$$Kt = \frac{\sigma_{ma}}{\sigma_{gross}} = \frac{20.086}{6.67} = 3.01$$

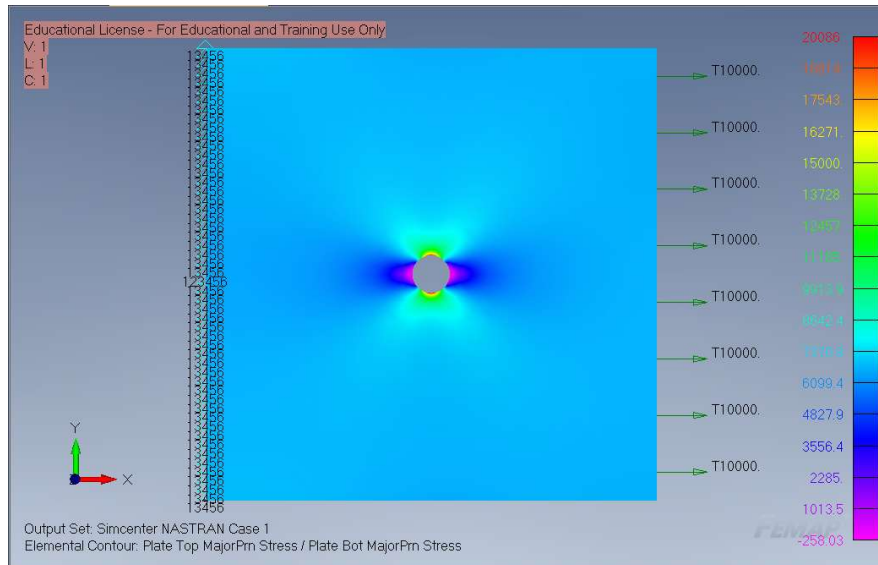


Figure 15: Femap Solution for Open Hole Tension Plate

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8.3 Python Solver

Figure 16 shows the Max Principal Stress fringe plot for the same mesh as used by FEMAP containing only CQ4 elements. The maximum principal stress is 20.373 ksi, 1.9% difference from the elasticity solution. This difference is within reasonable limits to consider the PySolver as validated for CQ4 elements. Figure 17 shows the same model with a course mesh of CQ4 elements and a refined region of CQ8 elements around the hole area of interest with CQ7 elements spanning the transition boundary. The results of this model shown in Figure 18 produced a maximum principal stress of 19.347 ksi. Considering the coarseness of this mesh an error 3.26% is reasonable to consider this solver validated for CQ8 and CQ7 elements. Refining the mesh further in the transition region did improve correlation with a stress of 19.624 ksi and an error of 1.8%

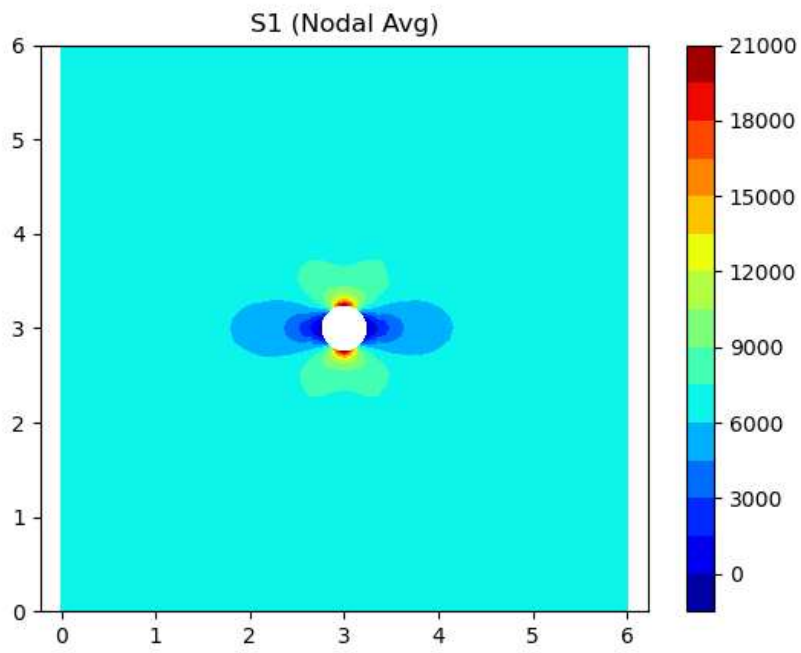


Figure 16: Python Solver for Open Hole Plate

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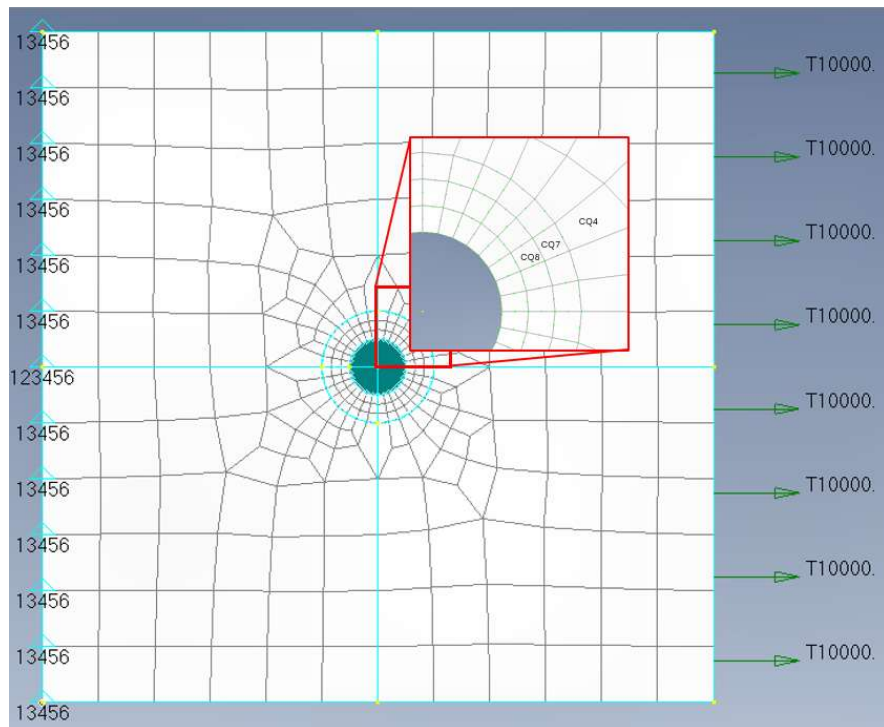


Figure 17: CQ8/CQ7 Open Hole Plate Model

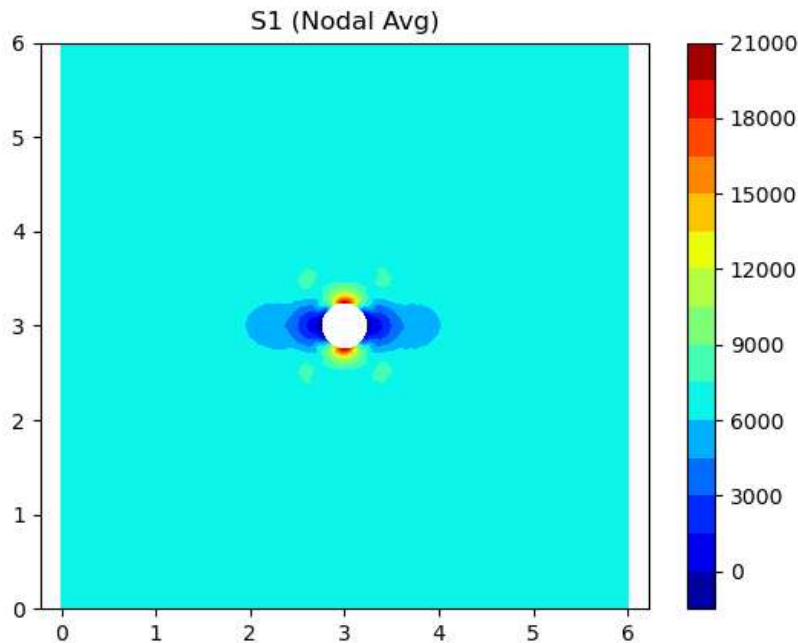


Figure 18: CQ8/CQ7 Open Hole Plate Results

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9 Kt Solution from the Initial Problem

From the initial problem statement the goal is to determine the K_t of the plate that does not properly match what is found in [2].

The plate was subjected to a 1000 lbf of tension load. Fixed in the lower left corner node with x displacements fixed at the left edge. It is assumed that the surrounding structure of the vertical frame is sufficiently stiff to eliminate displacement in the y direction. Thus all y displacements are fixed along the bottom edge of the plate.

The plate is re-meshed with increasing fidelity until stress convergence is achieved. This is done for a model containing only CQUAD4 elements in the critical area and one containing only CQUAD8 elements in the critical area. For the CQUAD8 model, the CQUAD7 and CQUAD6 elements are used as transition elements between the CQUAD8 and CQUAD4 elements. This transition is made outside of the critical area.